

Newman-Penrose Formalism Calculations

Stephani et al., Equation (22.29)

Metric

$$ds^2 = -w(\rho)^2 dt \otimes dt + \frac{e^{(-b(\rho))}}{E(\rho)h(\rho)^2} d\rho \otimes d\rho + \frac{1}{h(\rho)^2} dy \otimes dy + \frac{\rho}{h(\rho)^2} dy \otimes d\phi + \frac{\rho}{h(\rho)^2} d\phi \otimes dy + \left(\frac{\rho^2 + E(\rho)}{h(\rho)^2} \right) d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, \rho, y, \phi)$$

Metric Functions

$$E(\rho) = a\rho - \rho^2 + c$$

$$w^2(\rho) = e^{(-b(\rho))} h(\rho)^2$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} w(\rho) \right) dt + \left(-\frac{\sqrt{2} e^{(-\frac{1}{2} b(\rho))}}{2 \sqrt{E(\rho)} h(\rho)} \right) d\rho$$

$$n = \left(\frac{1}{2} \sqrt{2} w(\rho) \right) dt + \left(\frac{\sqrt{2} e^{(-\frac{1}{2} b(\rho))}}{2 \sqrt{E(\rho)} h(\rho)} \right) d\rho$$

$$m = \left(\frac{\sqrt{2}}{2 h(\rho)} \right) dy + \left(\frac{\sqrt{2} \rho}{2 h(\rho)} - \frac{i \sqrt{2} \sqrt{E(\rho)}}{2 h(\rho)} \right) d\phi$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2h(\rho)} \right) dy + \left(\frac{\sqrt{2}\rho}{2h(\rho)} + \frac{i\sqrt{2}\sqrt{E(\rho)}}{2h(\rho)} \right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2w(\rho)} \right) dt + \left(-\frac{1}{2} \sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}h(\rho) \right) d\rho$$

$$n = \left(-\frac{\sqrt{2}}{2w(\rho)} \right) dt + \left(\frac{1}{2} \sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}h(\rho) \right) d\rho$$

$$m = \left(\frac{i\sqrt{2}\rho h(\rho)}{2\sqrt{E(\rho)}} + \frac{1}{2} \sqrt{2}h(\rho) \right) dy + \left(-\frac{i\sqrt{2}h(\rho)}{2\sqrt{E(\rho)}} \right) d\phi$$

$$\overline{m} = \left(-\frac{i\sqrt{2}\rho h(\rho)}{2\sqrt{E(\rho)}} + \frac{1}{2} \sqrt{2}h(\rho) \right) dy + \left(\frac{i\sqrt{2}h(\rho)}{2\sqrt{E(\rho)}} \right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2w(\rho)} \partial_t - \frac{1}{2} \sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}h(\rho) \partial_\rho$$

$$\Delta = -\frac{\sqrt{2}}{2w(\rho)} \partial_t + \frac{1}{2} \sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}h(\rho) \partial_\rho$$

$$\delta = \frac{i\sqrt{2}\rho h(\rho)}{2\sqrt{E(\rho)}} + \frac{1}{2} \sqrt{2}h(\rho) \partial_y - \frac{i\sqrt{2}h(\rho)}{2\sqrt{E(\rho)}} \partial_\phi$$

$$\overline{\delta} = -\frac{i\sqrt{2}\rho h(\rho)}{2\sqrt{E(\rho)}} + \frac{1}{2} \sqrt{2}h(\rho) \partial_y + \frac{i\sqrt{2}h(\rho)}{2\sqrt{E(\rho)}} \partial_\phi$$

Spin Coefficients

$$\rho = \frac{\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)\frac{\partial}{\partial\rho}E(\rho)}{8\sqrt{E(\rho)}} - \frac{1}{2}\sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}\frac{\partial}{\partial\rho}h(\rho)$$

$$\sigma = -\frac{\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)\frac{\partial}{\partial\rho}E(\rho)}{8\sqrt{E(\rho)}} - \frac{1}{4}i\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)$$

$$\kappa = 0$$

$$\mu = \frac{\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)\frac{\partial}{\partial\rho}E(\rho)}{8\sqrt{E(\rho)}} - \frac{1}{2}\sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}\frac{\partial}{\partial\rho}h(\rho)$$

$$\lambda = -\frac{\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)\frac{\partial}{\partial\rho}E(\rho)}{8\sqrt{E(\rho)}} + \frac{1}{4}i\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)$$

$$\nu = 0$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = -\frac{\sqrt{2}\sqrt{E(\rho)}e^{(\frac{1}{2}b(\rho))}h(\rho)\frac{\partial}{\partial\rho}w(\rho)}{4w(\rho)} - \frac{1}{8}i\sqrt{2}e^{(\frac{1}{2}b(\rho))}h(\rho)$$

$$\gamma = -\frac{\sqrt{2}\sqrt{E(\rho)}e^{\left(\frac{1}{2}b(\rho)\right)}h(\rho)\frac{\partial}{\partial\rho}w(\rho)}{4w(\rho)} + \frac{1}{8}i\sqrt{2}e^{\left(\frac{1}{2}b(\rho)\right)}h(\rho)$$

Ricci Tensor Components

$$\begin{aligned}\Phi_{00} = & -\frac{1}{16}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}b(\rho) + \frac{1}{4}E(\rho)e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}b(\rho)\frac{\partial}{\partial\rho}h(\rho) - \frac{1}{8}e^{b(\rho)}h(\rho)^2\frac{\partial^2}{(\partial\rho)^2}E(\rho) + \frac{3}{8}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}h(\rho) \\ & - \frac{1}{2}E(\rho)e^{b(\rho)}\frac{\partial}{\partial\rho}h(\rho)^2 + \frac{1}{2}E(\rho)e^{b(\rho)}h(\rho)\frac{\partial^2}{(\partial\rho)^2}h(\rho) + \frac{e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}w(\rho)}{8w(\rho)} - \frac{E(\rho)e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}h(\rho)\frac{\partial}{\partial\rho}w(\rho)}{2w(\rho)} - \frac{1}{8}e^{b(\rho)}h(\rho)^2\end{aligned}$$

$$\Phi_{01} = 0$$

$$\begin{aligned}\Phi_{02} = & \frac{1}{16}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}b(\rho) + \frac{1}{8}i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}b(\rho) + \frac{1}{8}e^{b(\rho)}h(\rho)^2\frac{\partial^2}{(\partial\rho)^2}E(\rho) - \frac{1}{8}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}h(\rho) \\ & + \frac{e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}w(\rho)}{8w(\rho)} - \frac{1}{4}i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}h(\rho) + \frac{i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}w(\rho)}{4w(\rho)} + \frac{1}{4}e^{b(\rho)}h(\rho)^2\end{aligned}$$

$$\Phi_{10} = 0$$

$$\begin{aligned}\Phi_{11} = & \frac{E(\rho)e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}b(\rho)\frac{\partial}{\partial\rho}w(\rho)}{8w(\rho)} + \frac{1}{8}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}h(\rho) - \frac{1}{4}E(\rho)e^{b(\rho)}\frac{\partial}{\partial\rho}h(\rho)^2 \\ & + \frac{e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}w(\rho)}{8w(\rho)} + \frac{E(\rho)e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}h(\rho)\frac{\partial}{\partial\rho}w(\rho)}{4w(\rho)} + \frac{E(\rho)e^{b(\rho)}h(\rho)^2\frac{\partial^2}{(\partial\rho)^2}w(\rho)}{4w(\rho)} + \frac{1}{16}e^{b(\rho)}h(\rho)^2\end{aligned}$$

$$\Phi_{12} = 0$$

$$\begin{aligned}\Phi_{20} = & \frac{1}{16}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}b(\rho) - \frac{1}{8}i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}b(\rho) + \frac{1}{8}e^{b(\rho)}h(\rho)^2\frac{\partial^2}{(\partial\rho)^2}E(\rho) - \frac{1}{8}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}h(\rho) \\ & + \frac{e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}E(\rho)\frac{\partial}{\partial\rho}w(\rho)}{8w(\rho)} + \frac{1}{4}i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)\frac{\partial}{\partial\rho}h(\rho) - \frac{i\sqrt{E(\rho)}e^{b(\rho)}h(\rho)^2\frac{\partial}{\partial\rho}w(\rho)}{4w(\rho)} + \frac{1}{4}e^{b(\rho)}h(\rho)^2\end{aligned}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "*I*"